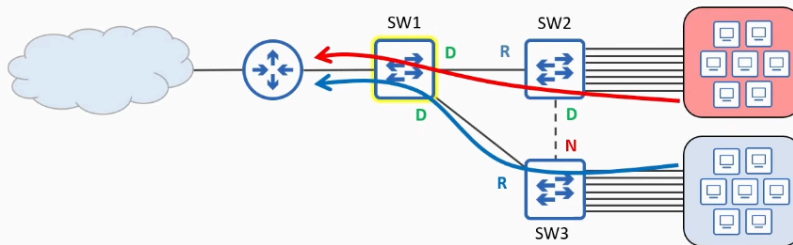


Day 21 (Part 5): Root Guard (STP Toolkit)

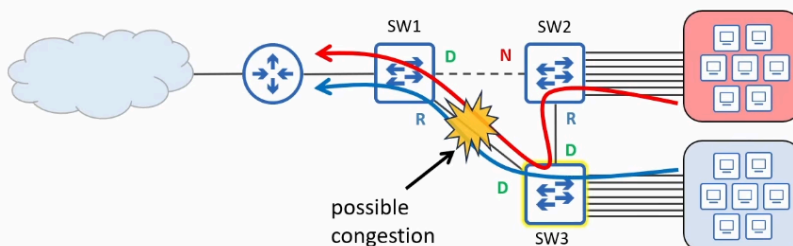
Root Bridge placement



- STP prevents loops by electing a root bridge and ensuring that each other switch has only one valid path to reach it.
- You shouldn't randomly select the root bridge. Some things you should consider include:
 - Optimal traffic flow
 - minimize latency
 - minimize congestion
 - Stability and reliability

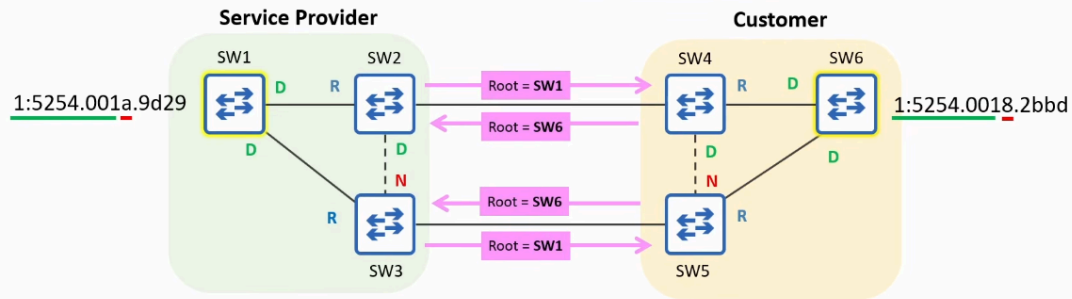
If SW3 takes the Root Bridge

Root Bridge placement





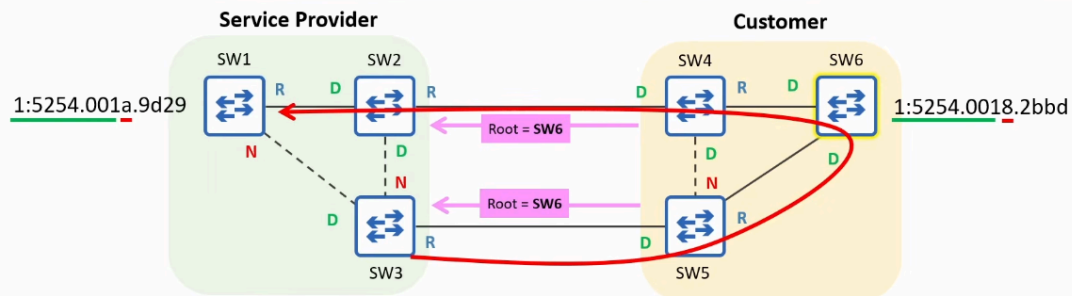
Root Guard: the problem



- Within your own LAN, you can easily control the root bridge by setting its priority to **0**.
 - But there are cases where you might connect your LAN to other switches outside of your direct control:
 - A service provider offering **Metro Ethernet** service to customers
 - often used to connect sites within a MAN (Metropolitan Area Network)
- Even if you set your root bridge's priority to 0, its role can be taken by another switch with a lower MAC address.



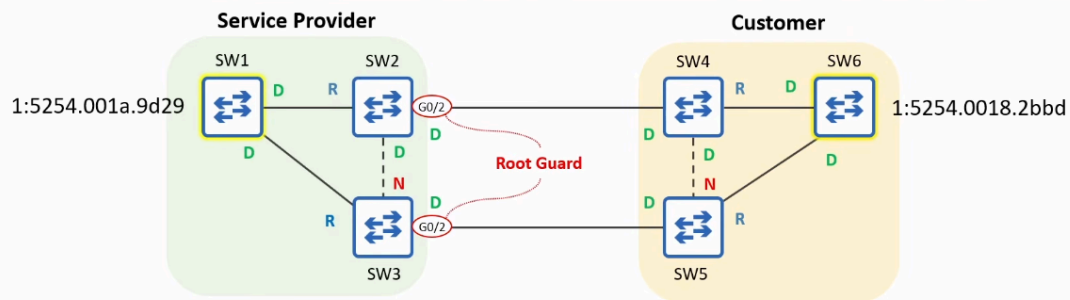
Root Guard: the problem



- With no safeguard in place, SW1, SW2, and SW3 accept SW6 as the root bridge, affecting the service provider's STP topology.
 - Frames from SW3 to SW1 must take a detour through the customer's LAN.
- **Root Guard** can be configured to protect your STP topology by preventing your switches from accepting **superior BPDUs** from switches outside of your control.
 - **Superior BPDUs** = a BPDU that is superior in the STP algorithm (e.g. claiming a better root bridge ID).



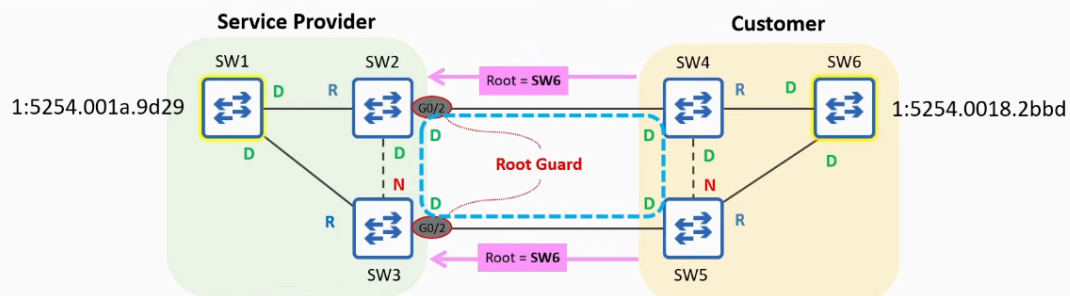
Root Guard: the solution



- If you want to ensure that the root bridge remains in your LAN, you can configure **Root Guard** on the ports connected to switches outside of your control.
 - SW2 G0/20, SW3 G0/20
- Use SW2(config-if)# **spanning-tree guard root** to enable Root Guard on a port.
 - There is no command to enable it by default from global config mode.



Root Guard: the solution



- If you want to ensure that the root bridge remains in your LAN, you can configure **Root Guard** on the ports connected to switches outside of your control.
 - SW2 G0/20, SW3 G0/20
- Use SW2(config-if)# **spanning-tree guard root** to enable Root Guard on a port.
 - There is no command to enable it by default from global config mode.
- If a Root Guard-enabled port receives a BPDU, it will enter the **Broken (Root Inconsistent)** state, effectively disabling it.
 - The port will not be able to forward data frames and will discard any frames it receives.
 - SW1, SW2, and SW3 won't accept SW6 as the root bridge.

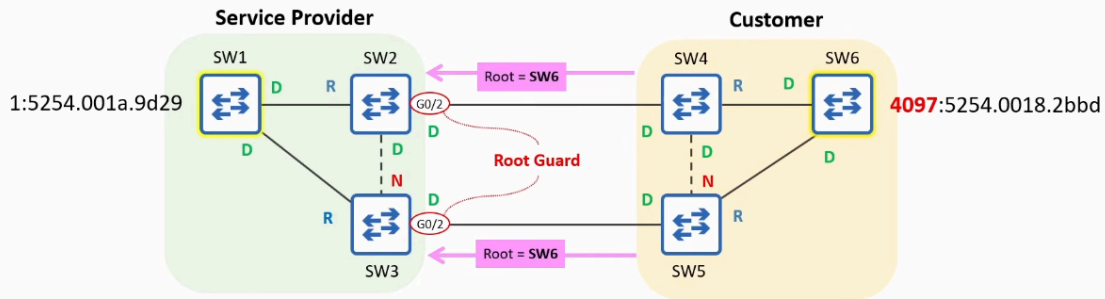


In this special case, SW2 - SW3 - SW4 - SW5 will have 2 DESIGNATED PORT
REMEMBER:

- There should be 1 DESIGNATED PORT per link in STP



Root Guard: the solution

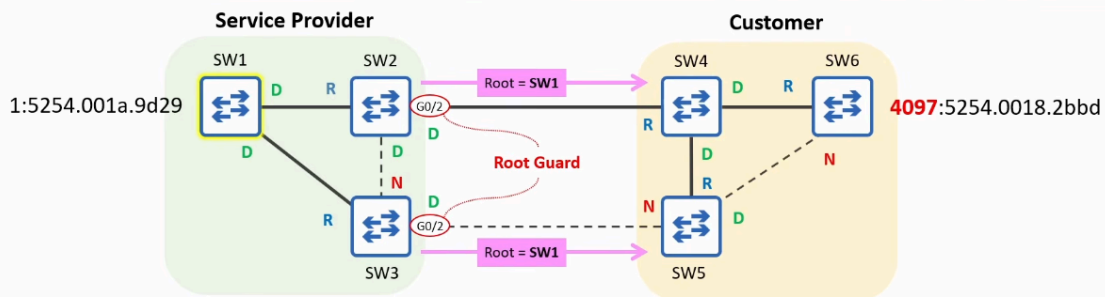


- To re-enable a port disabled by Root Guard, you must solve the issue that disabled the port.
 - The disabled port must stop receiving superior BPDUs.
 - Tell the customer to increase the priority value of their switch.
- Once the superior BPDUs received by SW2 G0/2 and SW3 G0/3 age out, the ports will automatically be re-enabled.
 - A BPDU's Max Age is 20 seconds by default.

Then CUSTOMER's LAN will take SW1 as a Root Bridge

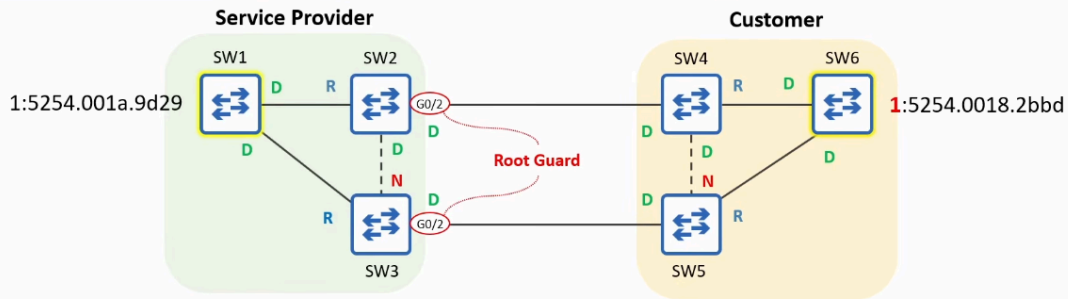


Root Guard: the solution





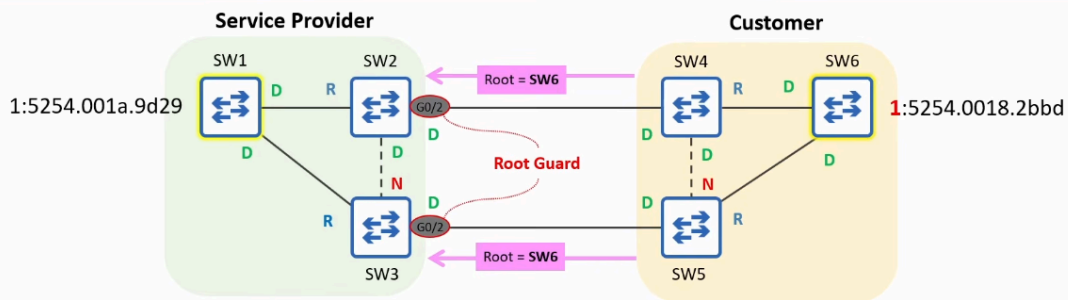
Root Guard: CLI demonstration



```
SW2(config)# interface g0/2
SW2(config-if)# spanning-tree guard root
*Sep 21 08:38:56.314: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port GigabitEthernet0/2.
```



Root Guard: CLI demonstration



```
SW2(config)# interface g0/2
SW2(config-if)# spanning-tree guard root
*Sep 21 08:38:56.314: %SPANTREE-2-ROOTGUARD_CONFIG_CHANGE: Root guard enabled on port GigabitEthernet0/2.
*Sep 21 08:38:56.321: %SPANTREE-2-ROOTGUARD_BLOCK: Root guard blocking port GigabitEthernet0/2 on VLAN0001.
SW2(config-if)# do show spanning-tree
!output omitted
Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Root FWD 4         128.1    P2p
Gi0/1          Desg FWD 4         128.2    P2p
Gi0/2          Desg BKN*4 128.3     P2p *ROOT_Inc
```

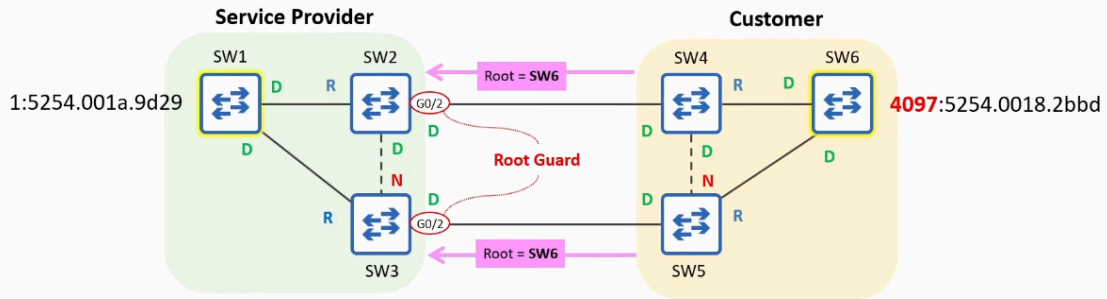
BKN = Broken
ROOT_Inc = Root Inconsistent

- Broken: can not forward or receive data frame
- Root inconsistent: tell us why port is broken

Change SW6 priority to 4097 and after 20 SECONDS, the interface is unblocked



Root Guard: CLI demonstration

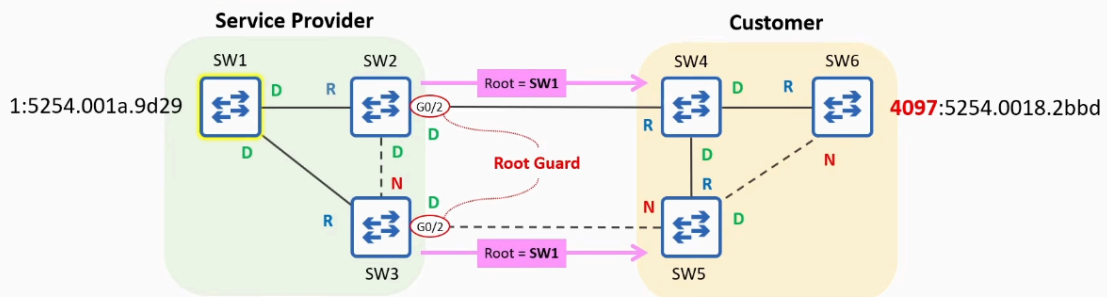


```
*Sep 21 08:54:26.955: %SPANTREE-2-ROOTGUARD_UNBLOCK: Root guard unblocking port GigabitEthernet0/2 on VLAN0001.  
SW2(config-if)# do show spanning-tree  
!output omitted  
Interface      Role Sts Cost      Prio.Nbr Type  
-----  
Gi0/0          Root FWD 4         128.1    P2p  
Gi0/1          Desg FWD 4         128.2    P2p  
Gi0/2          Desg FWD 4         128.3    P2p
```

Then Customer's LAN will take SW1 as Root Bridge



Root Guard: CLI demonstration





Summary

- When selecting a LAN's root bridge, you should consider the following:
 - Optimal traffic flow
 - minimize latency
 - minimize congestion
 - Stability and reliability
- Within your own LAN, you can easily control the root bridge by setting its priority to 0.
 - There are cases where you might connect your switches to other switches outside of your control (e.g. service provider + client).
- **Root Guard** can be configured on specific ports to prevent them from accepting **superior BPDUs** from switches outside of your control.
- Use SW(config-if)# **spanning-tree guard root** to enable Root Guard on a port.
 - There is no command to enable it by default from global config mode.
- Root Guard prevents a port from becoming a root port if it receives a superior BPDU.
 - If the port receives a superior BPDU, it becomes Broken (BKN) / Root Inconsistent (ROOT_Inc).
- If the port stops receiving superior BPDUs, it will automatically recover.